

Week 9: Neural Network

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Motivation (1/2)

Given a dataset D ,

x_1	x_2
0	1
2	3
4	5

How many coefficients θ_j are there for a polynomial of degree p ?
 $\binom{n}{r} = \frac{n!}{r!(n-r)!}$, where n is the sum of the number of attributes and polynomial degrees, and r is the number of polynomial degrees.

Motivation (2/2)

How about this dataset,

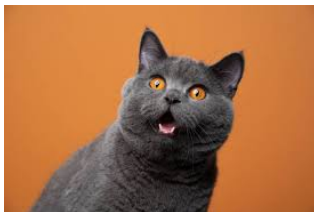


Figure: A cat W100xH100 istockphoto 1361394182

How many coefficients θ_j are there for a polynomial of degree p ?

Artificial Neural Network

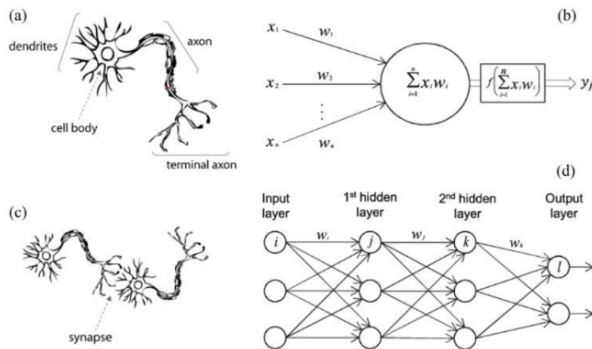


Figure 3. A biological neuron in comparison to an artificial neural network: (a) human neuron; (b) artificial neuron; (c) biological synapse; and (d) ANN synapses [39].

Figure: Artificial Neural Network

Artificial Neural Network

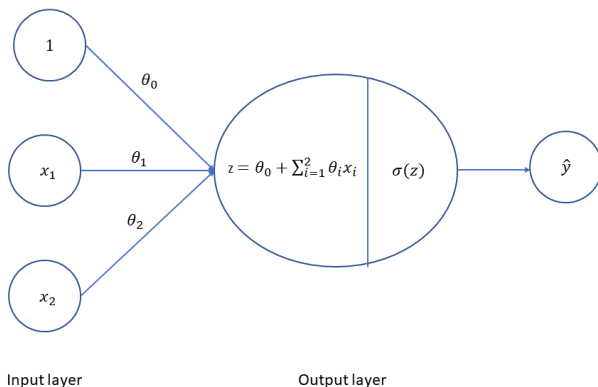
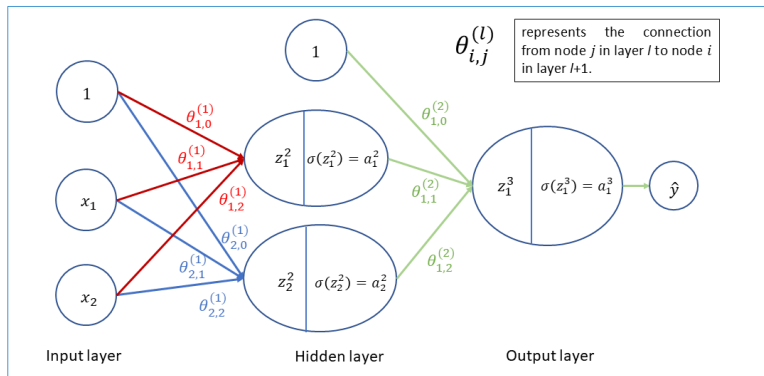


Figure: A simple Neural Network: Perceptron

Artificial Neural Network



$$a_1^2 = \sigma(\theta_{1,0}^{(1)} + \theta_{1,1}^{(1)}x_1 + \theta_{1,2}^{(1)}x_2)$$

$$a_2^2 = \sigma(\theta_{2,0}^{(1)} + \theta_{2,1}^{(1)}x_1 + \theta_{2,2}^{(1)}x_2)$$

$$a_1^3 = \sigma(\theta_{1,0}^{(2)} + \theta_{1,1}^{(2)}a_1^2 + \theta_{1,2}^{(2)}a_2^2)$$

$$\theta^{(1)} = \begin{bmatrix} \theta_{1,0}^{(1)} & \theta_{1,1}^{(1)} & \theta_{1,2}^{(1)} \\ \theta_{2,0}^{(1)} & \theta_{2,1}^{(1)} & \theta_{2,2}^{(1)} \end{bmatrix} \quad \theta^{(2)} = \begin{bmatrix} \theta_{1,0}^{(2)} & \theta_{1,1}^{(2)} & \theta_{1,2}^{(2)} \end{bmatrix}$$

Figure: Multilayer Perceptron

Training Data Example

AND Operation

x_1	x_2	y
0	0	0
0	1	0
1	0	0
1	1	1

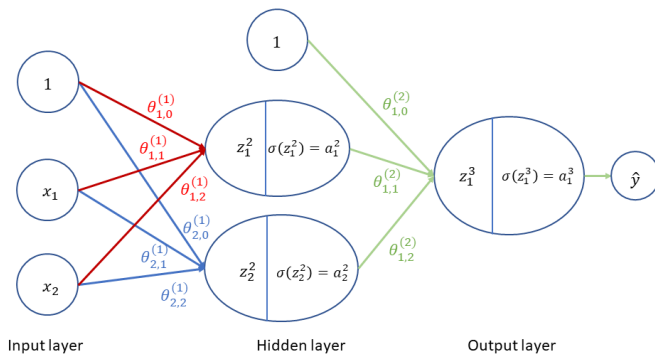
XNOR Operation

x_1	x_2	y
0	0	1
0	1	0
1	0	0
1	1	1

Training Process

- **Cost Function:** $J(\theta) = -y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})$
- **Goal:** $\theta^* = \arg \min J(\theta)$
- **For each sample in the training data:**
 - Step 1: Forward pass
Calculate $\hat{y}$, i.e. a^L , where a big L is the output layer.
 - Step 2: Calculate loss of the output layer, i.e. $a^L - y$.
 - Step 3: Backward pass (Backpropagation)
Find $\frac{\partial J(\theta)}{\partial \theta_{ij}}$
 - Step 4: Update each $\theta_{i,j}^{(l)}$ (weights)

Step 1&2: Forward pass & Error of last layer



Step 3: Backward pass (Backpropagation) 1/6

$\theta^{(2)}$

- $$\frac{\partial J(\theta)}{\partial \theta_{1,0}^{(2)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial \theta_{1,0}^{(2)}}$$

- $$\frac{\partial J(\theta)}{\partial \theta_{1,1}^{(2)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial \theta_{1,1}^{(2)}}$$

- $$\frac{\partial J(\theta)}{\partial \theta_{1,2}^{(2)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial \theta_{1,2}^{(2)}}$$

Step 3: Backward pass (Backpropagation) 2/6

$\theta^{(2)}$, e.g., $\theta_{1,0}^{(2)}$

$$\bullet \quad \frac{\partial J(\theta)}{\partial \theta_{1,0}^{(2)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial \theta_{1,0}^{(2)}}$$

$$\bullet \quad \frac{\partial J(\theta)}{\partial a_1^3} = \frac{\partial (-y \log(a_1^3) - (1-y) \log(1-a_1^3))}{\partial a_1^3} = -\frac{y}{a_1^3} + \frac{1-y}{1-a_1^3} = \frac{1-y}{1-a_1^3} - \frac{y}{a_1^3}$$

$$\bullet \quad \frac{\partial a_1^3}{\partial z_1^3} = \frac{\partial \sigma(z_1^3)}{\partial z_1^3} = \sigma(z_1^3) \cdot (1 - \sigma(z_1^3)) = a_1^3 \cdot (1 - a_1^3)$$

$$\bullet \quad \frac{\partial z_1^3}{\partial \theta_{1,0}^{(2)}} = \frac{\partial (\theta_{1,0}^{(2)} + \theta_{1,1}^{(2)} \cdot a_1^2 + \theta_{1,2}^{(2)} \cdot a_2^2)}{\partial \theta_{1,0}^{(2)}} = 1$$

$$\bullet \quad \text{So that, } \frac{\partial J(\theta)}{\partial \theta_{1,0}^{(2)}} = \left(\frac{1-y}{1-a_1^3} - \frac{y}{a_1^3} \right) \cdot (a_1^3 \cdot (1 - a_1^3)) \cdot 1 \\ = (a_1^3 - y) \cdot 1$$

Step 3: Backward pass (Backpropagation) 3/6

- $\frac{\partial J(\theta)}{\partial \theta_{1,0}^{(2)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial \theta_{1,0}^{(2)}} = (a_1^3 - y) \cdot 1$
- $\frac{\partial J(\theta)}{\partial \theta_{1,1}^{(2)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial \theta_{1,1}^{(2)}} = (a_1^3 - y) \cdot a_1^2$
- $\frac{\partial J(\theta)}{\partial \theta_{1,2}^{(2)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial \theta_{1,2}^{(2)}} = (a_1^3 - y) \cdot a_2^2$
- Let define $\delta_1^3 = (a_1^3 - y)$ be the error of the first node in the last layer ($L = 3$ in this example).
- We then get, $\frac{\partial J(\theta)}{\partial \theta_{i,j}^{(l)}} = \delta_i^{l+1} \cdot a_j^l$

Step 3: Backward pass (Backpropagation) 4/6

$\theta^{(1)}$

$$\bullet \quad \frac{\partial J(\theta)}{\partial \theta_{1,0}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_1^2} \cdot \frac{\partial a_1^2}{\partial z_1^2} \cdot \frac{\partial z_1^2}{\partial \theta_{1,0}^{(1)}}$$

$$\bullet \quad \frac{\partial J(\theta)}{\partial \theta_{1,1}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_1^2} \cdot \frac{\partial a_1^2}{\partial z_1^2} \cdot \frac{\partial z_1^2}{\partial \theta_{1,1}^{(1)}}$$

$$\bullet \quad \frac{\partial J(\theta)}{\partial \theta_{1,2}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_1^2} \cdot \frac{\partial a_1^2}{\partial z_1^2} \cdot \frac{\partial z_1^2}{\partial \theta_{1,2}^{(1)}}$$

$$\bullet \quad \frac{\partial J(\theta)}{\partial \theta_{2,0}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_2^2} \cdot \frac{\partial a_2^2}{\partial z_2^2} \cdot \frac{\partial z_2^2}{\partial \theta_{2,0}^{(1)}}$$

$$\bullet \quad \frac{\partial J(\theta)}{\partial \theta_{2,1}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_2^2} \cdot \frac{\partial a_2^2}{\partial z_2^2} \cdot \frac{\partial z_2^2}{\partial \theta_{2,1}^{(1)}}$$

$$\bullet \quad \frac{\partial J(\theta)}{\partial \theta_{2,2}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_2^2} \cdot \frac{\partial a_2^2}{\partial z_2^2} \cdot \frac{\partial z_2^2}{\partial \theta_{2,2}^{(1)}}$$

Step 3: Backward pass (Backpropagation) 5/6

$\theta^{(1)}$

- $\bullet \frac{\partial J(\theta)}{\partial \theta_{1,0}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_1^2} \cdot \frac{\partial a_1^2}{\partial z_1^2} \cdot \frac{\partial z_1^2}{\partial \theta_{1,0}^{(1)}} = (a_1^3 - y) \cdot \theta_{1,1}^{(2)} \cdot a_1^2(1 - a_1^2) \cdot 1$
- $\bullet \frac{\partial J(\theta)}{\partial \theta_{1,1}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_1^2} \cdot \frac{\partial a_1^2}{\partial z_1^2} \cdot \frac{\partial z_1^2}{\partial \theta_{1,1}^{(1)}} = (a_1^3 - y) \cdot \theta_{1,1}^{(2)} \cdot a_1^2(1 - a_1^2) \cdot a_1^1$
- $\bullet \frac{\partial J(\theta)}{\partial \theta_{1,2}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_1^2} \cdot \frac{\partial a_1^2}{\partial z_1^2} \cdot \frac{\partial z_1^2}{\partial \theta_{1,2}^{(1)}} = (a_1^3 - y) \cdot \theta_{1,1}^{(2)} \cdot a_1^2(1 - a_1^2) \cdot a_2^1$
- $\bullet \frac{\partial J(\theta)}{\partial \theta_{2,0}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_2^2} \cdot \frac{\partial a_2^2}{\partial z_2^2} \cdot \frac{\partial z_2^2}{\partial \theta_{2,0}^{(1)}} = (a_1^3 - y) \cdot \theta_{1,2}^{(2)} \cdot a_2^2(1 - a_2^2) \cdot 1$
- $\bullet \frac{\partial J(\theta)}{\partial \theta_{2,1}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_2^2} \cdot \frac{\partial a_2^2}{\partial z_2^2} \cdot \frac{\partial z_2^2}{\partial \theta_{2,1}^{(1)}} = (a_1^3 - y) \cdot \theta_{1,2}^{(2)} \cdot a_2^2(1 - a_2^2) \cdot a_1^1$
- $\bullet \frac{\partial J(\theta)}{\partial \theta_{2,2}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_2^2} \cdot \frac{\partial a_2^2}{\partial z_2^2} \cdot \frac{\partial z_2^2}{\partial \theta_{2,2}^{(1)}} = (a_1^3 - y) \cdot \theta_{1,2}^{(2)} \cdot a_2^2(1 - a_2^2) \cdot a_2^1$

Note that a_1^1 is x_1 , a_2^1 is x_2

Step 3: Backward pass (Backpropagation) 6/6

- let define $\delta_i^l = \delta_i^{l+1} \cdot \theta_{i,j}^{(l)} \cdot a_j^l \cdot (1 - a_j^l)$, where $l = L - 1, L - 2, \dots, 2$.

Based on the observation above, we can conclude that

- $\delta_1^2 = \delta_1^3 \cdot \theta_{1,1}^{(2)} \cdot a_1^2(1 - a_1^2)$
- $\delta_2^2 = \delta_1^3 \cdot \theta_{1,2}^{(2)} \cdot a_2^2(1 - a_2^2)$
- We still get, $\frac{\partial J(\theta)}{\partial \theta_{i,j}^{(l)}} = \delta_i^{l+1} \cdot a_j^l$, with a condition that
 - if $l = L$ then $\delta_i^l = (a_j^l - y)$
 - else if $l = L - 1, L - 2, \dots, 2$ then $\delta_i^l = \delta_i^{l+1} \cdot \theta_{i,j}^{(l)} \cdot a_j^l \cdot (1 - a_j^l)$

Step 4: Update each $\theta_{i,j}^{(l)}$ (weights)

- $\theta_{i,j}^{(l)} = \theta_{i,j}^{(l)} - \alpha \cdot \frac{\partial J(\theta)}{\partial \theta_{i,j}^{(l)}}$
- $\theta_{i,j}^{(l)} = \theta_{i,j}^{(l)} - \alpha \cdot (\delta_i^{l+1} \cdot a_j^l)$, where
 - $\theta_{i,j}^{(l)}$: The parameter (e.g., weight) connecting node (j) in layer (l) to node (i) in layer $l + 1$.
 - α : The learning rate.
 - δ_i^{l+1} : The error term (delta) for node (i) in layer $l + 1$.
 - a_j^l : The activation of node (j) in layer (l).

Activation Functions

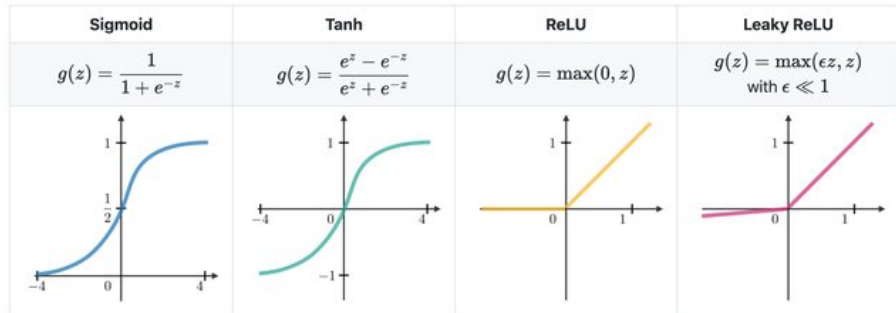


Figure: Activation function Ref:

<http://dx.doi.org/10.13140/RG.2.2.14692.60800>

Softmax Functions

Multi-Class Classification with NN and SoftMax Function

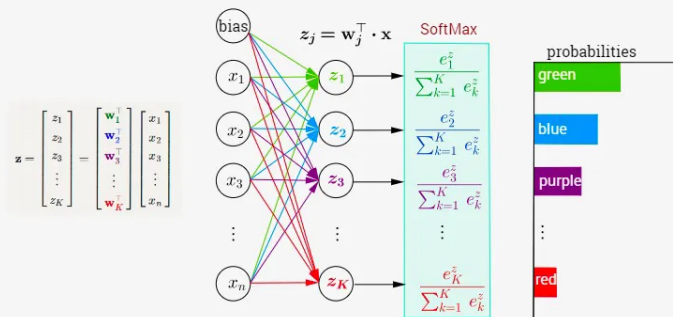
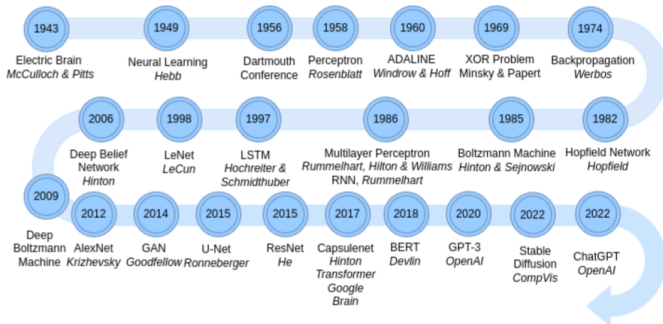


Figure: Softmax function Ref:

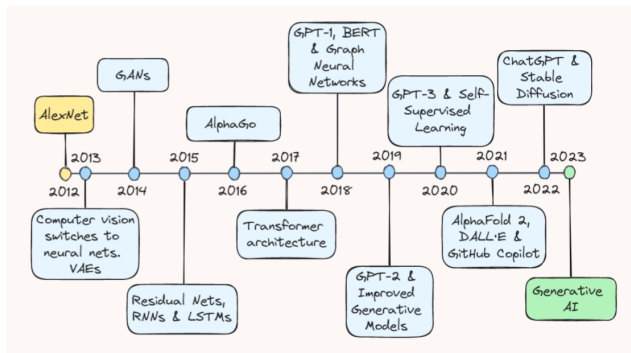
<https://adeveloperdiary.com/data-science/deep-learning/neural-network-with-softmax-in-python/>

History of Neural Networks (1/2)



Ref: <https://pub.towardsai.net/a-brief-history-of-neural-nets-472107bc2c9c>

History of Neural Networks (2/2)



Neural network zoo:

<https://www.asimovinstitute.org/neural-network-zoo/>

- Abiodun, O.I., et. al.(2018).State-of-the-art in artificial neural network applications: A survey Heliyon, 4(11).